Light and Electron Microscopes: What Each One Can Show

Name _____ Period ____

Module 2 · Cell Theory and Cell Types · LT 2.1 and LT 2.2

Predict first. You have a classroom microscope and a slice of onion skin. Name one thing inside the cell you expect to see, and one thing you think is too small. Check yourself at the end.

Two kinds of microscope

	Light microscope	Electron microscope
What makes the image	Visible light bent by glass lenses	A beam of electrons focused by magnets
How much it magnifies	About 40× to 1,000× (classroom: 40×, 100×, 400×)	Up to about 1,000,000× and more
Smallest detail it can separate	About 0.2 µm, so cells and the larger organelles	Below 1 nm, so ribosomes, membranes and even large molecules
Living or dead?	Living cells can be watched moving and dividing	Dead only: the sample sits in a vacuum and is fixed and coated
Color	True color, or stains you add	Black and white; any color is added on a computer afterward
Where you find it	Every school lab	Research labs and hospitals; costly and room-sized

Electron microscopes come in two kinds. A **transmission electron microscope (TEM)** sends electrons through a very thin slice, so it shows the inside of a cell: ribosomes, the folds of the endoplasmic reticulum, the inner membrane of a mitochondrion. A **scanning electron microscope (SEM)** bounces electrons off the surface, so it shows the outside of a cell or an insect's eye in sharp, three-dimensional detail.

What each one can show

With a light microscope at 400× you can see the cell wall, the cell membrane at the edge of the cell, the cytoplasm, the nucleus, chloroplasts and the large central vacuole in a plant cell. Total magnification is the eyepiece times the objective: a 10× eyepiece with a 40× objective gives 400×.

Only an electron microscope shows ribosomes, the detail of the endoplasmic reticulum and Golgi apparatus, the two layers of a membrane, and the inside of a mitochondrion. Bacteria show up as small dots or rods under a light microscope; their inner parts need an electron microscope.

Better tools, stronger theory

- **1665.** Robert Hooke's microscope shows the empty walls of dead cork cells. He calls them cells.
- **1670s.** Anton van Leeuwenhoek's stronger single lens shows living single-celled organisms swimming in pond water.
- **1838–1839.** After looking at many plant and animal tissues, scientists propose that all living things are made of cells.
- **1855.** Watching cells divide adds the idea that cells come only from existing cells.
- **1930s–1950s.** The electron microscope reveals ribosomes, the endoplasmic reticulum and the inside of mitochondria. Cells across every kingdom share the same basic parts, and the modern cell theory adds tenets: metabolism happens inside cells, DNA passes from cell to cell, and all cells have the same basic chemical makeup.

Each new tool gave new evidence, and the theory grew to fit it. That is how science works: theories are strong because of evidence, and they are revised when better tools find more.

Look back: what the Unit 2 test asks you to do

1. **LT 2.1, nature of science.** Explain how a new tool, such as the electron microscope, led to a deeper understanding of cells and a revised cell theory.
2. **LT 2.1, evidence.** Read an observation made with a microscope and name the cell-theory tenet it supports best.
3. **LT 2.1, history.** Put the discoveries in order and say what each tool made possible.
4. **LT 2.2, plant and animal cells.** Name what you could see in a plant cell under a light microscope that you would not see in an animal cell: the cell wall, chloroplasts and a large central vacuole.
5. **LT 2.2, prokaryotes and eukaryotes.** Know that ribosomes are in both kinds of cell and that it took an electron microscope to show them.

Quick check

1. A student uses a 10× eyepiece and a 4× objective. What is the total magnification? (A) 14× (B) 40× (C) 400× (D) 4×
2. Which structure can be seen clearly only with an electron microscope? (A) Nucleus (B) Cell wall (C) Ribosome (D) Chloroplast
3. A scientist wants to watch a living amoeba change shape. Which tool should she use, and why? (A) An electron microscope, because it magnifies more (B) A light microscope, because the sample can stay alive (C) An electron microscope, because it shows color (D) A light microscope, because it shows ribosomes
4. Under a light microscope, a pond alga cell and an amoeba are compared. What could be seen in the alga but not the amoeba? (A) A nucleus and cytoplasm (B) A cell membrane (C) A cell wall and chloroplasts (D) Ribosomes
5. Electron microscopes showed that bacteria, plants and animals all have ribosomes. Which tenet of the modern cell theory does this support best? (A) Cells come only from existing cells (B) All cells have the same basic chemical makeup (C) Only animals are made of cells (D) Theories never change

Answers: 1 B · 2 C · 3 B · 4 C · 5 B. Back to your prediction: the nucleus and cell wall are easy to see in onion skin; ribosomes are far too small.

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